

Higher Education and Research English

Instructor guide for advanced ESL learners working in higher education and research

Audience: faculty, postdoctoral researchers, graduate students, lab managers, research administrators, grant staff, ethics-board coordinators, and academic program leaders

Focus: A higher education and research English curriculum for grant proposals, lab meetings, peer review, research ethics, authorship, data management, academic presentations, and institutional collaboration.

Designed for advanced ESL learners who already use professional English and need industry-specific terminology, realistic meetings, role-play pressure, careful pushback, and polished workplace outputs.

Teaching stance: this is language and workplace-communication training, not legal, medical, financial, safety, or regulatory advice. Instructors should connect every scenario to the learner's current company policies, local rules, and approved procedures.

Purpose and Course Logic

A higher education and research English curriculum for grant proposals, lab meetings, peer review, research ethics, authorship, data management, academic presentations, and institutional collaboration.

Core language challenge

Advanced learners do not only need vocabulary. They need the ability to ask which standard applies, who owns the decision, what evidence is sufficient, what risk is being accepted, and how to disagree without sounding vague, defensive, or reckless.

Each module trains a realistic workplace pressure point with role-specific terms, decision language, pushback practice, and bounded decision activities learners can apply to their own work.

Course objectives

- Use higher education and research terminology accurately in meetings, written updates, handoffs, escalations, reviews, and client or stakeholder conversations.
- Turn vague requests into specific questions about evidence, owner, deadline, constraint, risk, and decision rights.
- Push back on unsafe, unsupported, noncompliant, unrealistic, or poorly scoped proposals while preserving professional trust.
- Handle realistic dialogues from the field, including conflict, uncertainty, documentation gaps, customer or stakeholder pressure, and cross-functional disagreement.
- Select language that produces concise workplace outputs: briefing notes, escalation updates, meeting scripts, risk memos, decision records, and follow-up messages.

Instructor Module Plans

Module 1. Research Questions and Study Design (90 minutes)

Frame research claims with scope and methodological discipline.

Learners should be able to

- Use these terms accurately: research question, hypothesis, methodology, limitation.
- Explain the workplace tension: Hypothesis, design, sample, limitations, and inference must be aligned.
- Respond professionally when a stakeholder says: Use stronger language to attract attention.
- Select the evidence, tradeoff, owner, and decision required for a study-design caveat.

Customized scenario

Workplace pressure

A lab wants to describe an exploratory study as definitive.

Use stronger language to attract attention.

Hypothesis, design, sample, limitations, and inference must be aligned.

Classroom sequence

1. Terminology selection: distinguish the term that names the decision variable from three plausible alternatives.
2. Risk triage: select the stakeholder, decision, evidence gap, operating constraint, and cost of being wrong from a bounded scenario set.
3. Pushback ladder: choose the strongest sequence from clarification to evidence-based objection to consequence to decision request.

4. Decision artifact check: select the facts, caveat, owner, and next action that belong in a study-design caveat.

Module 2. Grant Proposals and Specific Aims (90 minutes)

Write aims that are ambitious but testable.

Learners should be able to

- Use these terms accurately: specific aims, significance, innovation, feasibility.
- Explain the workplace tension: Feasibility, significance, innovation, approach, and milestones need balance.
- Respond professionally when a stakeholder says: Keep every aim to look comprehensive.
- Select the evidence, tradeoff, owner, and decision required for a specific-aims revision.

Customized scenario

Workplace pressure

A proposal includes too many objectives for the budget.

Keep every aim to look comprehensive.

Feasibility, significance, innovation, approach, and milestones need balance.

Classroom sequence

1. Terminology selection: distinguish the term that names the decision variable from three plausible alternatives.
2. Risk triage: select the stakeholder, decision, evidence gap, operating constraint, and cost of being wrong from a bounded scenario set.
3. Pushback ladder: choose the strongest sequence from clarification to evidence-based objection to consequence to decision request.
4. Decision artifact check: select the facts, caveat, owner, and next action that belong in a specific-aims revision.

Module 3. Lab Meetings and Data Challenges (90 minutes)

Question data respectfully but rigorously.

Learners should be able to

- Use these terms accurately: control, replication, protocol drift, analysis assumption.
- Explain the workplace tension: Controls, replication, protocol drift, and analysis assumptions need review.
- Respond professionally when a stakeholder says: Say the data are bad.
- Select the evidence, tradeoff, owner, and decision required for a lab-meeting question set.

Customized scenario

Workplace pressure

A student presents inconsistent results.

Say the data are bad.

Controls, replication, protocol drift, and analysis assumptions need review.

Classroom sequence

1. Terminology selection: distinguish the term that names the decision variable from three plausible alternatives.
2. Risk triage: select the stakeholder, decision, evidence gap, operating constraint, and cost of being wrong from a bounded scenario set.

3. Pushback ladder: choose the strongest sequence from clarification to evidence-based objection to consequence to decision request.
4. Decision artifact check: select the facts, caveat, owner, and next action that belong in a lab-meeting question set.

Module 4. Research Ethics and Human Subjects (90 minutes)

Set boundaries around consent, risk, and protocol adherence.

Learners should be able to

- Use these terms accurately: IRB, informed consent, secondary use, human subjects.
- Explain the workplace tension: Consent, IRB approval, privacy, and secondary-use rules may apply.
- Respond professionally when a stakeholder says: Analyze it because the data already exist.
- Select the evidence, tradeoff, owner, and decision required for a ethics consultation note.

Customized scenario

Workplace pressure

A researcher wants to use data for a new question outside the approved protocol.

Analyze it because the data already exist.

Consent, IRB approval, privacy, and secondary-use rules may apply.

Classroom sequence

1. Terminology selection: distinguish the term that names the decision variable from three plausible alternatives.
2. Risk triage: select the stakeholder, decision, evidence gap, operating constraint, and cost of being wrong from a bounded scenario set.
3. Pushback ladder: choose the strongest sequence from clarification to evidence-based objection to consequence to decision request.
4. Decision artifact check: select the facts, caveat, owner, and next action that belong in a ethics consultation note.

Module 5. Authorship, Collaboration, and Credit (90 minutes)

Discuss contribution and authorship early.

Learners should be able to

- Use these terms accurately: authorship, contribution, corresponding author, acknowledgment.
- Explain the workplace tension: Contribution, criteria, order, acknowledgments, and publication norms must be explicit.
- Respond professionally when a stakeholder says: Agree to avoid conflict.
- Select the evidence, tradeoff, owner, and decision required for a authorship agreement draft.

Customized scenario

Workplace pressure

A collaborator expects authorship after a small advisory role.

Agree to avoid conflict.

Contribution, criteria, order, acknowledgments, and publication norms must be explicit.

Classroom sequence

1. Terminology selection: distinguish the term that names the decision variable from three plausible alternatives.

2. Risk triage: select the stakeholder, decision, evidence gap, operating constraint, and cost of being wrong from a bounded scenario set.
3. Pushback ladder: choose the strongest sequence from clarification to evidence-based objection to consequence to decision request.
4. Decision artifact check: select the facts, caveat, owner, and next action that belong in a authorship agreement draft.

Module 6. Peer Review and Revision Responses (90 minutes)

Respond to criticism without defensiveness.

Learners should be able to

- Use these terms accurately: peer review, major revision, response letter, scope.
- Explain the workplace tension: Tone, evidence, scope, feasibility, and transparent limitations matter.
- Respond professionally when a stakeholder says: Reject the comment sharply.
- Select the evidence, tradeoff, owner, and decision required for a reviewer-response paragraph.

Customized scenario

Workplace pressure

Reviewers ask for additional analyses outside the original scope.

Reject the comment sharply.

Tone, evidence, scope, feasibility, and transparent limitations matter.

Classroom sequence

1. Terminology selection: distinguish the term that names the decision variable from three plausible alternatives.
2. Risk triage: select the stakeholder, decision, evidence gap, operating constraint, and cost of being wrong from a bounded scenario set.
3. Pushback ladder: choose the strongest sequence from clarification to evidence-based objection to consequence to decision request.
4. Decision artifact check: select the facts, caveat, owner, and next action that belong in a reviewer-response paragraph.

Module 7. Data Management and Reproducibility (90 minutes)

Explain data stewardship and reproducible workflows.

Learners should be able to

- Use these terms accurately: metadata, repository, provenance, reproducibility.
- Explain the workplace tension: Metadata, code, provenance, privacy, and repository requirements need attention.
- Respond professionally when a stakeholder says: Upload it anyway.
- Select the evidence, tradeoff, owner, and decision required for a data-management checklist.

Customized scenario

Workplace pressure

A dataset lacks clear metadata before publication.

Upload it anyway.

Metadata, code, provenance, privacy, and repository requirements need attention.

Classroom sequence

1. Terminology selection: distinguish the term that names the decision variable from three plausible alternatives.
2. Risk triage: select the stakeholder, decision, evidence gap, operating constraint, and cost of being wrong from a bounded scenario set.
3. Pushback ladder: choose the strongest sequence from clarification to evidence-based objection to consequence to decision request.
4. Decision artifact check: select the facts, caveat, owner, and next action that belong in a data-management checklist.

Module 8. Academic Presentations and Conferences (90 minutes)

Present claims with confidence and caveats.

Learners should be able to

- Use these terms accurately: generalizability, limitation, future work, conference Q&A.
- Explain the workplace tension: Population, context, method, and future work should be addressed calmly.
- Respond professionally when a stakeholder says: Defend every conclusion.
- Select the evidence, tradeoff, owner, and decision required for a conference Q&A response.

Customized scenario

Workplace pressure

A conference audience challenges the study's generalizability.

Defend every conclusion.

Population, context, method, and future work should be addressed calmly.

Classroom sequence

1. Terminology selection: distinguish the term that names the decision variable from three plausible alternatives.
2. Risk triage: select the stakeholder, decision, evidence gap, operating constraint, and cost of being wrong from a bounded scenario set.
3. Pushback ladder: choose the strongest sequence from clarification to evidence-based objection to consequence to decision request.
4. Decision artifact check: select the facts, caveat, owner, and next action that belong in a conference Q&A response.

Nomenclature and Jargon

These are classroom working definitions. Learners should adapt wording to their organization's policies, systems, and local regulatory environment.

Research Questions and Study Design

Term	Working meaning	Collocations	Contrast and workplace line
research question	In higher education and research, research question is a field-specific concept used to locate the relevant fact, boundary, or decision variable. Use it when teams need to frame research claims with scope and methodological discipline.	clarify research question; document research question	Contrast: research question should be defined against the relevant evidence and decision boundary, rather than used as a vague label. Example: Before we proceed, please clarify research question and the evidence that supports it.

Term	Working meaning	Collocations	Contrast and workplace line
hypothesis	In higher education and research, hypothesis is a field-specific concept used to locate the relevant fact, boundary, or decision variable. Use it when teams need to frame research claims with scope and methodological discipline.	clarify hypothesis; document hypothesis	Contrast: hypothesis should be defined against the relevant evidence and decision boundary, rather than used as a vague label. Example: Before we proceed, please clarify hypothesis and the evidence that supports it.
methodology	In higher education and research, methodology is information that can be checked to support a claim, decision, investigation, or compliance conclusion. Use it when teams need to frame research claims with scope and methodological discipline.	clarify methodology; document methodology	Contrast: methodology should be defined against the relevant evidence and decision boundary, rather than used as a vague label. Example: Before we proceed, please clarify methodology and the evidence that supports it.
limitation	In higher education and research, limitation is a field-specific concept used to locate the relevant fact, boundary, or decision variable. Use it when teams need to frame research claims with scope and methodological discipline.	clarify limitation; document limitation	Contrast: limitation should be defined against the relevant evidence and decision boundary, rather than used as a vague label. Example: Before we proceed, please clarify limitation and the evidence that supports it.

Grant Proposals and Specific Aims

Term	Working meaning	Collocations	Contrast and workplace line
specific aims	In higher education and research, specific aims is a field-specific concept used to locate the relevant fact, boundary, or decision variable. Use it when teams need to write aims that are ambitious but testable.	clarify specific aims; document specific aims	Contrast: specific aims should be defined against the relevant evidence and decision boundary, rather than used as a vague label. Example: Before we proceed, please clarify specific aims and the evidence that supports it.
significance	In higher education and research, significance is a field-specific concept used to locate the relevant fact, boundary, or decision variable. Use it when teams need to write aims that are ambitious but testable.	clarify significance; document significance	Contrast: significance should be defined against the relevant evidence and decision boundary, rather than used as a vague label. Example: Before we proceed, please clarify significance and the evidence that supports it.
innovation	In higher education and research, innovation is a field-specific concept used to locate the relevant fact, boundary, or decision variable. Use it when teams need to write aims that are ambitious but testable.	clarify innovation; document innovation	Contrast: innovation should be defined against the relevant evidence and decision boundary, rather than used as a vague label. Example: Before we proceed, please clarify innovation and the evidence that supports it.
feasibility	In higher education and research, feasibility is a field-specific concept used to locate the relevant fact, boundary, or decision variable. Use it when teams need to write aims that are ambitious but testable.	clarify feasibility; document feasibility	Contrast: feasibility should be defined against the relevant evidence and decision boundary, rather than used as a vague label. Example: Before we proceed, please clarify feasibility and the evidence that supports it.

Lab Meetings and Data Challenges

Term	Working meaning	Collocations	Contrast and workplace line
control	A process, approval, check, or technical safeguard designed to reduce risk.	clarify control; document control	Contrast: control should be defined against the relevant evidence and decision boundary, rather than used as a vague label. Example: Before we proceed, please clarify control and the evidence that supports it.
replication	In higher education and research, replication is a field-specific concept used to locate the relevant fact, boundary, or decision variable. Use it when teams need to question data respectfully but rigorously.	clarify replication; document replication	Contrast: replication should be defined against the relevant evidence and decision boundary, rather than used as a vague label. Example: Before we proceed, please clarify replication and the evidence that supports it.
protocol drift	In higher education and research, protocol drift is an approved rule or decision framework that sets what is permitted, required, or acceptable. Use it when teams need to question data respectfully but rigorously.	apply the protocol drift; comply with the protocol drift	Contrast: protocol drift is an approved rule or threshold, not an informal preference. Example: Please confirm which protocol drift applies before we commit the team.
analysis assumption	In higher education and research, analysis assumption is a structured examination of evidence, process, or results used to reach a defensible conclusion. Use it when teams need to question data respectfully but rigorously.	conduct a analysis assumption; document the analysis assumption	Contrast: analysis assumption is a structured examination of evidence; it is not a conclusion reached before the evidence is checked. Example: The team will conduct a analysis assumption before it recommends a corrective action.

Research Ethics and Human Subjects

Term	Working meaning	Collocations	Contrast and workplace line
IRB	In higher education and research, IRB is a field-specific concept used to locate the relevant fact, boundary, or decision variable. Use it when teams need to set boundaries around consent, risk, and protocol adherence.	clarify IRB; document IRB	Contrast: IRB should be defined against the relevant evidence and decision boundary, rather than used as a vague label. Example: Before we proceed, please clarify IRB and the evidence that supports it.
informed consent	In higher education and research, informed consent is a field-specific concept used to locate the relevant fact, boundary, or decision variable. Use it when teams need to set boundaries around consent, risk, and protocol adherence.	clarify informed consent; document informed consent	Contrast: informed consent should be defined against the relevant evidence and decision boundary, rather than used as a vague label. Example: Before we proceed, please clarify informed consent and the evidence that supports it.
secondary use	In higher education and research, secondary use is a field-specific concept used to locate the relevant fact, boundary, or decision variable. Use it when teams need to set boundaries around consent, risk, and protocol adherence.	clarify secondary use; document secondary use	Contrast: secondary use should be defined against the relevant evidence and decision boundary, rather than used as a vague label. Example: Before we proceed, please clarify secondary use and the evidence that supports it.
human subjects	In higher education and research, human subjects is a field-specific concept used to locate the relevant fact, boundary, or decision variable. Use it when teams need to set boundaries around consent, risk, and protocol adherence.	clarify human subjects; document human subjects	Contrast: human subjects should be defined against the relevant evidence and decision boundary, rather than used as a vague label. Example: Before we proceed, please clarify human subjects and the evidence that supports it.

Authorship, Collaboration, and Credit

Term	Working meaning	Collocations	Contrast and workplace line
authorship	In higher education and research, authorship is a field-specific concept used to locate the relevant fact, boundary, or decision variable. Use it when teams need to discuss contribution and authorship early.	clarify authorship; document authorship	Contrast: authorship should be defined against the relevant evidence and decision boundary, rather than used as a vague label. Example: Before we proceed, please clarify authorship and the evidence that supports it.
contribution	In higher education and research, contribution is a field-specific concept used to locate the relevant fact, boundary, or decision variable. Use it when teams need to discuss contribution and authorship early.	clarify contribution; document contribution	Contrast: contribution should be defined against the relevant evidence and decision boundary, rather than used as a vague label. Example: Before we proceed, please clarify contribution and the evidence that supports it.
corresponding author	In higher education and research, corresponding author is a field-specific concept used to locate the relevant fact, boundary, or decision variable. Use it when teams need to discuss contribution and authorship early.	clarify corresponding author; document corresponding author	Contrast: corresponding author should be defined against the relevant evidence and decision boundary, rather than used as a vague label. Example: Before we proceed, please clarify corresponding author and the evidence that supports it.
acknowledgment	In higher education and research, acknowledgment is a field-specific concept used to locate the relevant fact, boundary, or decision variable. Use it when teams need to discuss contribution and authorship early.	clarify acknowledgment; document acknowledgment	Contrast: acknowledgment should be defined against the relevant evidence and decision boundary, rather than used as a vague label. Example: Before we proceed, please clarify acknowledgment and the evidence that supports it.

Peer Review and Revision Responses

Term	Working meaning	Collocations	Contrast and workplace line
peer review	In higher education and research, peer review is a structured examination of evidence, process, or results used to reach a defensible conclusion. Use it when teams need to respond to criticism without defensiveness.	conduct a peer review; document the peer review	Contrast: peer review is a structured examination of evidence; it is not a conclusion reached before the evidence is checked. Example: The team will conduct a peer review before it recommends a corrective action.
major revision	In higher education and research, major revision is a field-specific concept used to locate the relevant fact, boundary, or decision variable. Use it when teams need to respond to criticism without defensiveness.	clarify major revision; document major revision	Contrast: major revision should be defined against the relevant evidence and decision boundary, rather than used as a vague label. Example: Before we proceed, please clarify major revision and the evidence that supports it.
response letter	In higher education and research, response letter is a workplace communication artifact that records facts, interpretation, and the decision or action requested. Use it when teams need to respond to criticism without defensiveness.	prepare a response letter; circulate the response letter	Contrast: response letter is a reviewable communication artifact, not an undocumented verbal assurance. Example: The response letter should state the evidence, caveat, decision owner, and next action.
scope	Defined boundary of work, responsibility, deliverables, assumptions, and exclusions.	clarify scope; document scope	Contrast: scope should be defined against the relevant evidence and decision boundary, rather than used as a vague label. Example: Before we proceed, please clarify scope and the evidence that supports it.

Data Management and Reproducibility

Term	Working meaning	Collocations	Contrast and workplace line
metadata	In higher education and research, metadata is a defined information asset or analytical construct used to measure, explain, or predict an outcome. Use it when teams need to explain data stewardship and reproducible workflows.	clarify metadata; document metadata	Contrast: metadata should be defined against the relevant evidence and decision boundary, rather than used as a vague label. Example: Before we proceed, please clarify metadata and the evidence that supports it.
repository	In higher education and research, repository is a field-specific concept used to locate the relevant fact, boundary, or decision variable. Use it when teams need to explain data stewardship and reproducible workflows.	clarify repository; document repository	Contrast: repository should be defined against the relevant evidence and decision boundary, rather than used as a vague label. Example: Before we proceed, please clarify repository and the evidence that supports it.
provenance	In higher education and research, provenance is information that can be checked to support a claim, decision, investigation, or compliance conclusion. Use it when teams need to explain data stewardship and reproducible workflows.	clarify provenance; document provenance	Contrast: provenance should be defined against the relevant evidence and decision boundary, rather than used as a vague label. Example: Before we proceed, please clarify provenance and the evidence that supports it.
reproducibility	In higher education and research, reproducibility is a field-specific concept used to locate the relevant fact, boundary, or decision variable. Use it when teams need to explain data stewardship and reproducible workflows.	clarify reproducibility; document reproducibility	Contrast: reproducibility should be defined against the relevant evidence and decision boundary, rather than used as a vague label. Example: Before we proceed, please clarify reproducibility and the evidence that supports it.

Academic Presentations and Conferences

Term	Working meaning	Collocations	Contrast and workplace line
generalizability	In higher education and research, generalizability is a field-specific concept used to locate the relevant fact, boundary, or decision variable. Use it when teams need to present claims with confidence and caveats.	clarify generalizability; document generalizability	Contrast: generalizability should be defined against the relevant evidence and decision boundary, rather than used as a vague label. Example: Before we proceed, please clarify generalizability and the evidence that supports it.
limitation	In higher education and research, limitation is a field-specific concept used to locate the relevant fact, boundary, or decision variable. Use it when teams need to present claims with confidence and caveats.	clarify limitation; document limitation	Contrast: limitation should be defined against the relevant evidence and decision boundary, rather than used as a vague label. Example: Before we proceed, please clarify limitation and the evidence that supports it.
future work	In higher education and research, future work is a field-specific concept used to locate the relevant fact, boundary, or decision variable. Use it when teams need to present claims with confidence and caveats.	clarify future work; document future work	Contrast: future work should be defined against the relevant evidence and decision boundary, rather than used as a vague label. Example: Before we proceed, please clarify future work and the evidence that supports it.
conference Q&A	In higher education and research, conference Q&A is a field-specific concept used to locate the relevant fact, boundary, or decision variable. Use it when teams need to present claims with confidence and caveats.	clarify conference Q&A; document conference Q&A	Contrast: conference Q&A should be defined against the relevant evidence and decision boundary, rather than used as a vague label. Example: Before we proceed, please clarify conference Q&A and the evidence that supports it.

Industry-Specific Meeting Moves

Situation	Useful language
Research Questions and Study Design	Before we commit, I want to confirm research question, hypothesis, the owner, and the evidence behind the decision. If hypothesis, design, sample, limitations, and inference must be aligned., I recommend we document the risk and agree on the next step.
Grant Proposals and Specific Aims	Before we commit, I want to confirm specific aims, significance, the owner, and the evidence behind the decision. If feasibility, significance, innovation, approach, and milestones need balance., I recommend we document the risk and agree on the next step.
Lab Meetings and Data Challenges	Before we commit, I want to confirm control, replication, the owner, and the evidence behind the decision. If controls, replication, protocol drift, and analysis assumptions need review., I recommend we document the risk and agree on the next step.
Research Ethics and Human Subjects	Before we commit, I want to confirm IRB, informed consent, the owner, and the evidence behind the decision. If consent, irb approval, privacy, and secondary-use rules may apply., I recommend we document the risk and agree on the next step.
Authorship, Collaboration, and Credit	Before we commit, I want to confirm authorship, contribution, the owner, and the evidence behind the decision. If contribution, criteria, order, acknowledgments, and publication norms must be explicit., I recommend we document the risk and agree on the next step.
Peer Review and Revision Responses	Before we commit, I want to confirm peer review, major revision, the owner, and the evidence behind the decision. If tone, evidence, scope, feasibility, and transparent limitations matter., I recommend we document the risk and agree on the next step.
Data Management and Reproducibility	Before we commit, I want to confirm metadata, repository, the owner, and the evidence behind the decision. If metadata, code, provenance, privacy, and repository requirements need attention., I recommend we document the risk and agree on the next step.
Academic Presentations and Conferences	Before we commit, I want to confirm generalizability, limitation, the owner, and the evidence behind the decision. If population, context, method, and future work should be addressed calmly., I recommend we document the risk and agree on the next step.

High-pressure pushback frames

- I understand the urgency. The risk is that we move faster than the evidence or process supports.
- I am not blocking the goal. I am naming the condition we need before the decision is safe and credible.
- If we accept this risk, we should name the owner, document the assumption, and define the trigger for escalation.
- That may be possible, but not under the current scope, timeline, or approval path.
- Let's separate what we know, what we assume, and what still needs confirmation.

Assessment and Coaching

Performance rubric

Skill	Developing	Proficient	Strong
Terminology	Recognizes terms but uses them loosely.	Uses field terms accurately in context.	Defines terms, connects them to evidence, and explains decision impact.
Pushback	Disagrees vaguely or avoids disagreement.	Names concern with evidence and next step.	Balances urgency, relationship, risk, owner, and decision rights.
Scenario judgment	Focuses on one stakeholder's preference.	Identifies constraint, risk, and process.	Guides the group toward a documented, realistic decision.
Decision communication	Selects language without linking it to the decision.	Chooses language that names facts, owner, and next step.	Distinguishes evidence, tradeoff, authority, and escalation in a decision-ready response.

Source orientation

- Institutional research policies and IRB guidance.
- Grant-funder instructions and journal requirements.
- Field norms for authorship, data, and reproducibility.
- The learner's own company policies, SOPs, contracts, systems, templates, and approved communication standards.